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Project Name	ReliefOps QA(Disaster Relief & Volunteer Operations Questions-Answering Assistant)
Domain	Humanitarian Aid/Disaster Management
Problem Statement	During floods and humanitarian crises, NGO volunteers and community emergency responders must quickly verify Standard Operating Procedures (SOPs), evacuation shelter rules, and supply triage guidelines. However, critical operational instructions are buried across lengthy civil defense manuals and fragmented crisis datasets. Manual searching delays urgent logistical coordination, while general-purpose AI chatbots risk hallucinating unsafe emergency procedures. An AI engineering solution using Retrieval-Augmented Generation (RAG) and strict guardrails enables volunteers to get instant, accurate, and citation-backed operational answers 24/7.
Evidence of the Problem	Evidence 1: During active disasters, thousands of unstructured crisis reports, urgent needs, and volunteer coordination requests occur simultaneously, overwhelming human coordinators who must classify and cross-reference them against official relief SOPs. Source / citation: https://mimran.me/papers/HumAID_Human_Annotated_Disaster_Incidents_Data_from_Twitter_ICWSM21.pdf Evidence 2: National disaster management frameworks involve multi-agency coordination rules, temporary evacuation center (*Pusat Pemindahan Sementara* / PPS) registration procedures, and tiered disaster protocols that community volunteers struggle to navigate quickly when stored in static multi-page policy PDFs. Source / citation: https://www.researchgate.net/publication/348609721_Flood_Risk_Management_in_Malaysia_The_current_hindrances_for_flood_related_agencies
Initial Related Work (at least 3)	Study / solution 1: Manual SOP Handbooks & Static Government Disaster Portals Source / citation: https://www.ifrc.org/our-work/disasters-climate-and-crises Key limitation: Authoritative, but static and time-consuming for field volunteers to search across multiple PDFs during urgent mobilization Study / solution 2: Crisis Informatics Classification Benchmarks (HumAID & CrisisBench) Source / citation: https://crisisnlp.qcri.org/humaid_dataset.html Key limitation: Focuses only on classifying short crisis messages into categories rather than synthesizing step-by-step procedural QA answers from SOP documents. Study / solution 3: Retrieval-Augmented Generation (RAG) & Ungrounded General-Purpose LLMs Source / citation: https://proceedings.neurips.cc/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf Key limitation: Standard ungrounded LLMs lack access to localized disaster relief SOPs and frequently hallucinate unverified evacuation protocols, shelter rules, or emergency contact details unless constrained by a domain-specific RAG pipeline

Engineering Gap / Limitation	What practical or technical limitation remains unresolved? Gap / Limitation: **:** Existing crisis informatics solutions only perform static message classification without interactive procedural retrieval, while general-purpose LLMs lack strict grounding in approved civil defense SOPs and deterministic guardrails to prevent unsafe medical or life-safety advice.
Initial Project Objectives (2–3)	O1: To design a document-grounded AI Question-Answering assistant for disaster relief volunteers based on official emergency SOPs and verified crisis response datasets. O2: To develop the proposed RAG solution with semantic retrieval, source citations, and safety guardrails that refuse high-risk medical diagnosis or autonomous rescue dispatch queries. O3 (if applicable): To evaluate the solution objectively in terms of retrieval precision, answer groundedness (faithfulness), response latency, and safety guardrail refusal accuracy.
Target Users	NGO student volunteers, community relief coordinators, and logistics mobilization teams assisting in non-clinical disaster relief operations
Data Sources (at least 2)	Source 1: Human-Annotated Disaster Incidents & Volunteering Dataset — URL: https://huggingface.co/datasets/QCRI/HumAID-all Source 2: Humanitarian Crisis Response & Needs Dataset— URL: https://huggingface.co/datasets/QCRI/CrisisBench-all-lang Source 3 (if any): Comprehensive QA / RAG Benchmark Dataset & Public Disaster Relief SOP Documents (Hugging Face)-URL: https://huggingface.co/datasets/Successmove/comprehensive-qa-dataset
Similar to EduAdvisor	Both systems use a document-grounded Retrieval-Augmented Generation (RAG) architecture with input/output guardrails to provide instant, citation-backed answers from authoritative documents rather than relying on unverified general LLM memory.
Different from EduAdvisor	EduAdvisor operates in higher education academic advising to answer course registration and graduation questions, whereas ReliefOps QA operates in humanitarian disaster management—combining crisis informatics datasets (HumAID / CrisisBench) with disaster SOP manuals to guide volunteer logistics, shelter setup, and supply categorization under stricter safety boundaries.
Responsible AI Risks (at least 2)	**Risk 1: Hallucination of Operational Procedures** — The model invents an inaccurate shelter registration rule or supply handling protocol, causing logistical confusion among relief volunteers. **Risk 2: Safety / Out-of-Scope Medical & Rescue Triage** — A user asks for clinical medical diagnosis or life-or-death emergency rescue dispatch, leading to harmful reliance if the system does not immediately refuse and redirect to official emergency services (999). **Risk 3 (if any): Fairness & Regional Terminology Bias** — Global crisis datasets under-represent local Malaysian disaster terminology (e.g., *Pusat Pemindahan Sementara* / PPS), resulting in uneven retrieval quality across local vs. international SOP queries.